# Research status & verification

17 October 2026

Current classification and verification scope

> **Read these results and data with caution.**
> This is AI-generated research. The results, proofs, code and data have not been independently verified by human experts or formalized in Lean. Treat all conclusions as provisional and use them with caution. Automated checks establish only their stated computational scope.

## The current classification record

The targeted study records 35 decisions among 38 cases: 28 exclusions and seven unitary constructions. Three general categorification questions remain open: R6_4_155, R6_8_1660 and R8_2_24. A feasible set of necessary conditions is not a categorical construction, and a conditional obstruction is not a nonexistence proof.

| Rank | Bound | Rings | Excluded | Models | Open |
|---|---|---|---|---|---|
| 4 | 128 | 889,530 | 889,515 | 15 | 0 |
| 5 | 20 | 40,999 | 40,982 | 17 | 0 |
| 6 | 9 | 25,421 | 25,362 | 57 | 2 |
| 7 | 4 | 4,201 | 4,172 | 29 | 0 |
| 8 | 2 | 973 | 895 | 77 | 1 |

The bound is the largest multiplicity included in the displayed census, not necessarily a complete classification range. Rank six is complete through multiplicity three; its exact bands 5, 6, 7 and 9 are also complete, while bands 4 and 8 each contain one open case.

Rank eight is complete through multiplicity one, with exactly 48 Grothendieck rings. At exact multiplicity two it records 847 exclusions, 29 unitary models and one open case. There are therefore 29 or 30 general complex Grothendieck rings in that band.

Rank-six models comprise 56 unitary constructions and one spherical construction that is not pseudo-unitary. Completeness means coverage of the stated census by the project's recorded arguments; it does not assert independent human or formal verification.

Public atlas | GitHub repository

MATHEMATICAL FRONTIER

# Three open cases

Each question concerns general complex fusion categorification. The reductions below are necessary conditions with explicit limits.

## R6_4_155 / A general existence question

Neither a general complex categorification nor a general exclusion is established. The available reductions and necessary-condition calculations narrow the problem but do not construct an associator or supply a complete contradiction.

The applicable spherical reduction is Proposition 7.1 of the rank-six manuscript. Its use must remain distinct from the conclusion that a categorification exists.

## R6_8_1660 / Necessary center data do not settle categorification

The unrestricted sphericalization reduction in Section 7.5 of the rank-six manuscript is source-bound and has been replayed. It checks 640 signed quotients, 24 sign orbits and 58 nongauge character rejections.

A rank-37 Verlinde obstruction rejects a particular candidate witness, not the source fusion ring. Both existence and general nonexistence remain unresolved.

## R8_2_24 / An inner-square condition with a transport gap

Any complex categorification has a pseudo-unitary Galois conjugate whose rank-four neutral subcategory has 2D2 fusion rules. Conjugation by an odd invertible must fix all neutral simple labels and square tensor-isomorphically to conjugation by the neutral involution.

There is no unitary categorification, and a more general obstruction applies when the neutral category lies in the Galois class of the known unitary one. These scoped results do not settle the general complex case.

The candidate eight-slot coefficient argument requires four oriented B/C categorical transport identities in a common pivotal frame. The signs arising from the oriented crossing and the skew self-duality of the invertible object have not been established. The finite algebra and normalized unit contraction do not prove those transports. The conditional calculation contributes no classification decision.

> **Status discipline.** A restricted nonexistence result applies only under its stated assumptions. An exact computation of conditional equations does not discharge the missing categorical hypothesis.

EXACT COMPUTATIONAL EVIDENCE

# What was checked

## Ranks four and five

Both complete rank-four replays passed: 889,530 rings, 15 models and 889,515 exclusions. Each regenerated 6,470,006 signed orbits; an independent full unsigned recount also passed.

The reconstructed rank-five package passed all 40,999 pinned classes, 17 model matches and 40,982 general exclusions in normal and optimized Python. The two reconstructed ledgers agree byte-for-byte. The unavailable source ledger was not recovered; the reconstructed ledger has its own recorded digest.

## Ranks seven and eight

The complete base-certificate loops passed in normal Python. Separate source-bound transition replays and current-ledger checks passed in normal and optimized Python with exact parent restoration. These checks account for the current partitions displayed in this note.

The rank-eight base replay checks all 973 records, including 75 signed proof trees, four rank-seven subring proof transfers and six appendix tensor transfers. These finite checks do not establish the open R8_2_24 categorical transport identities.

## Rank six: a combined verification campaign

The complete initial process checked all 25,362 exclusion certificates: 25,355 passed and seven failed. Each of those seven then passed the complete unchanged checker under SymPy's Python backend. Exact matrix comparison attributes the failures to equation ordering or orientation and positional separating-vector conventions; no proof inequality or certificate was relaxed.

Both post-loop checks skipped by the initial failure gate passed separately. The model registry gives 57 matches, and the multiplicity-one partition gives 39 = 18 + 21. A hash-bound aggregate validates complete certificate coverage and 299 source/input bindings.

This is a combined campaign, not a successful single full run or an all-Python full-census pass. The initial process exited with failure, and that status is preserved in the evidence. Its loaded backend/version was not measured at startup; the global python-flint installation changed from 0.8.0 to 0.9.0 during the run, so worker and child processes may have loaded different versions.

The complete-replay orchestrator has passed identity and rejection-control checks but has not itself been executed end to end. Its existence must not be reported as an additional full-census success.

READING AND REPRODUCING THE RECORD

# Evidence and its limits

## Source identity

A checksum binds a statement or calculation to specified bytes. It cannot establish that their mathematics is correct.

## Exact replay

A program verifies its implemented finite equations, enumerations or tensor comparisons. Its mathematical assumptions remain part of the proof obligation.

## Categorical argument

A claimed construction or obstruction must prove the hypotheses connecting the finite calculation to a fusion category. An unproved transport identity is not supplied by an arithmetic PASS.

## Independent review

Separate AI review passes are not independent human verification. No human verification or Lean formalization is claimed here.

## Replay from the repository root

```
python3 tools/restore_data.py
python3 tools/verify_current_publication.py
python3 -O tools/verify_current_publication.py
```

Add --historical to run the complete source-bound certificate chain on isolated inputs. Checkers using assert require normal Python; an optimized invocation alone removes those assertions. The current-publication verifier also checks the identity and consistency of the published documents and their website copies.

The rank-specific full-census commands, dependencies, receipts and limitations are documented in the verification guide. The aggregate release checks are separate from the mathematical certificate replays.

## Current sources

**Research status**
**Verification guide**
**Public research status page**

The manuscript statements, individual ledger entries and their source-bound evidence should be read together. This note summarizes the current scope; it is not an independent proof certificate.